

What is a Method?

A method is a predefined block of code designed to perform a specific task. They prevent code duplication and make software modular.

```
static void myMethod() {  
    System.out.println("Modular task executed.");  
}
```

Parameters & Multiple Data

Information can be passed to methods as parameters. You can define as many as you need.

```
static void greet(String name, int age) {  
    System.out.println(name + " is " + age);  
}  
  
// Calling the method  
greet("Alice", 25);
```

Return Values

Instead of void, use a data type (int, double, etc.) to make a method send a value back.

```
static int square(int x) {  
    return x * x;  
}  
  
int result = square(5); // result is 25
```

Method Overloading

Multiple methods can have the **same name** as long as the number or type of parameters is different.

```
static int add(int a, int b) { return a + b; }  
static double add(double a, double b) { return a + b; }
```

Essential Summary

- **Static:** Means the method belongs to the class, not an object.
- **Void:** Signifies the method does not return values.
- **Reusability:** Write once, execute thousands of times.
- **Scope:** Variables declared inside a method are **local** to that method.